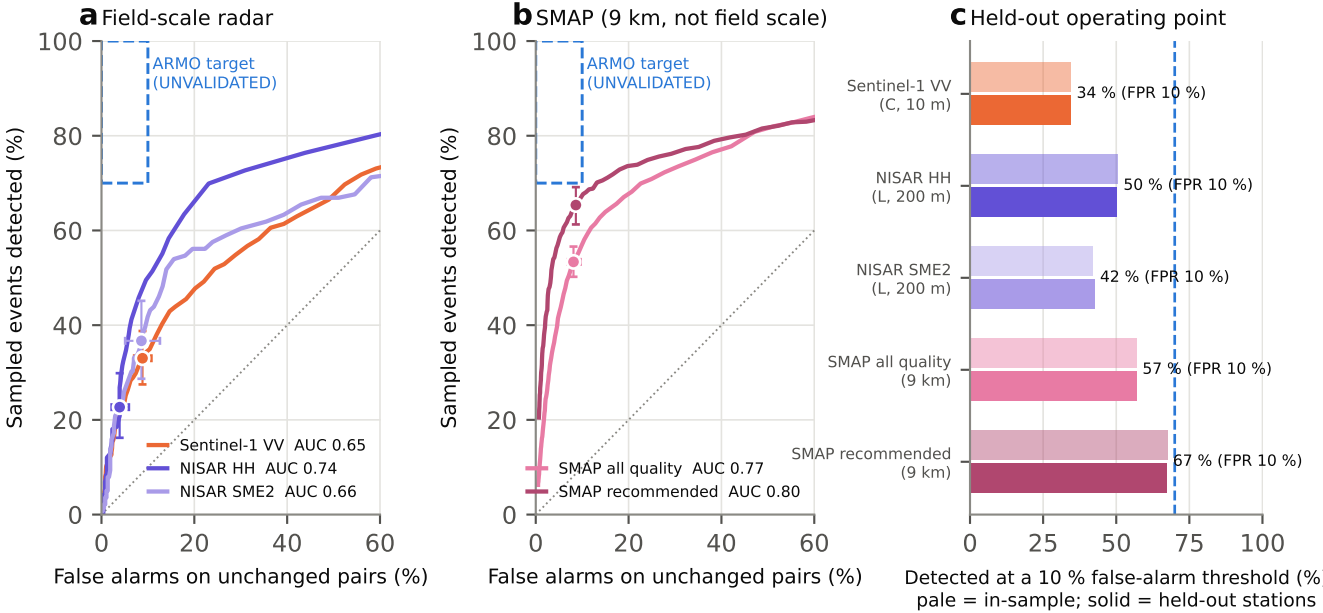

OBSERVED. Curves sweep the decision threshold in units of each sensor's no-change noise σ; dots = published 1σ rule with 95 % station-bootstrap CIs. Different sensors use different events and periods (S1 and SMAP 2025–2026; NISAR 2026, provisional SME2), so curves are not a same-cohort ranking. Held-out (c): threshold = 90th percentile of training-station negatives, scored on the other stations (5 folds). Dashed box = ARMO requirement (≥ 70 % detection at ≤ 10 % false alarms): a target, not a demonstrated result. The ≤ 0.038 m³/m³ noise requirement is MODEL-BASED (calibrated detection model), not measured here.